Mod

ϕ

$(d)(e) \mod phi = 1$

$d = 2753$

$(d)(17) \mod 3120 = 1$

$(2753)(17) \mod 3120 = 1$

$46,801 \mod 3120 = 1$

$1 = 1$

$$
\begin{array}{r}
15 \\
3120\overline{)46801} \\
31\,20 \\
\hline
15601 \\
15600 \\
\hline
1
\end{array}
$$

RSA Steps

1. Prime numbers : p & q
2. $n = p \cdot q$
3. $phi = (p-1)(q-1)$
4. $e \Rightarrow$ select e where

$$3 < e < phi \quad \text{And}$$

$$gcd(e, phi) = 1$$

(5) Private exponent "d"

Because $gcd(e, phi) = 1$

then

$$d e \equiv 1 \bmod phi$$

$$e * d \bmod phi = 1$$

⑥ RSA's main security relies on

$n = p * z$ ✓

$n = p * q$ ✗

Public Key (e, n)

$3 < e < phi$

Normally the public key is set to 65,537 if gcd(65537, phi) = 1

Private Key (d, n)

* Because gcd(e, phi) = 1

we can use an algorithm to select "d"

↳ EEA ⇒ Extended Euclidean Algorithm

Example 3

(1) $p = 11$

$q = 13$

(2) $n = p * q = 11 * 13 = 143$

(3) $\phi n = (p-1)(q-1) = 10 * 12 = 120$

(4) $e = ?$ ph

$3 < e < 120$

$\gcd(e, 120) = 1$

So, $\gcd(5, 120) = ? = 5$, $e = 5 \Rightarrow \times$

$\gcd(3, 120) = ? = 3$, $e = 3 \Rightarrow \times$

$\gcd(7, 120) = ? = 1$, $e = 7 \Rightarrow \checkmark$

(5) $d = ?$

EEA

$$\boxed{d \cdot e \bmod phi = 1}$$

$$\begin{array}{r} 6 \\ 120 \overline{)\ 721} \\ \underline{720} \\ 1 \end{array}$$

$$d = EEA(e, phi)$$

Prove that $\boxed{d = 103}$

$(103)(7) \bmod 120 = 1$

$721 \bmod 120 = 1$

$1 = 1$

if $m = 9$

Encrypt

$$c = m^e \bmod n = 9^7 \bmod 143$$

$$c = 48$$

Decrypt

$$m = c^d \bmod n = 48^{103} \bmod 143$$

$$m = 9$$